CNN Android Mobile Developer Code Challenge

Create, from scratch, an Android phone application that calls the Open Weather Map API and displays the results.

Items of Consideration

1. ***Do not*** use third party libraries in any form.
2. It is highly recommended that you write the app using Java
3. While the app is portrait only, be sure to handle all of the various sizes of phones
4. The application calls openweathermap.org. You will need to read their documentation to determine how to obtain a token to be used to call their API
5. The application does not need to get the location from the user or device. Use Atlanta, GA as the location

Data Service

Getting the Current Weather

Here is the query to get the current weather from openweathermap.org

```
http://api.openweathermap.org/data/2.5/weather?
q=Atlanta,ga&units=imperial
```

You will need to get an App ID for development from openweathermap.org. Visit http://openweathermap.org/appid to create a free account

Once you have an appId, you will need to append the App ID to the end of the url, like so:

```
http://api.openweathermap.org/data/2.5/weather?
q=Atlanta,ga&units=imperial&appid=123456
```

where *123456* is the App ID

The response for the current weather call will resemble this:


```
{
  "coord": {
    "lon": -84.39,
    "lat": 33.75
  },
  "weather": [
    {
      "id": 803,
      "main": "Clouds",
      "description": "broken clouds",
      "icon": "04d"
    }
  ],
  "base": "cmc stations",
  "main": {
    "temp": 64.83,
    "pressure": 1019,
    "humidity": 72,
    "temp_min": 62.01,
    "temp_max": 68
  },
  "wind": {
    "speed": 5.82,
    "deg": 200
  },
  "clouds": {
    "all": 75
  },
  "dt": 1461591863,
  "sys": {
    "type": 1,
    "id": 748,
    "message": 0.0044,
    "country": "US",
    "sunrise": 1461581639,
```

```
    "sunset": 1461629830
  },
  "id": 4180439,
  "name": "Atlanta",
  "cod": 200
}
```


> NOTE: A definition of each field can be found under the ***Weather parameters in API respond*** section here: http://openweathermap.org/current

Getting the 5 Day Forecast

Here is the query to get the 5 day forecast from openweathermap.org. Like the current weather, you will need to use your App ID

```
http://api.openweathermap.org/data/2.5/forecast/daily?
q=Atlanta,ga&units=imperial&cnt=5
```

The response for the 5-day forecast weather call will resemble this:


```
{
  "city": {
    "id": 4180439,
    "name": "Atlanta",
    "coord": {
      "lon": -84.387978,
      "lat": 33.749001
    },
    "country": "US",
    "population": 0
  },
  "cod": "200",
  "message": 0.0602,
  "cnt": 5,
  "list": [
    {
      "dt": 1461344400,
      "temp": {
        "day": 66.67,
        "min": 59.14,
        "max": 70.93,
```

```
      "night": 59.14,
      "eve": 66.56,
      "morn": 66.67
    },
    "pressure": 989.4,
    "humidity": 100,
    "weather": [
      {
        "id": 502,
        "main": "Rain",
        "description": "heavy intensity rain",
        "icon": "10d"
      }
    ],
    "speed": 3.58,
    "deg": 249,
    "clouds": 92,
    "rain": 12.7
  },
  {
    "dt": 1461430800,
    "temp": {
      "day": 71.87,
      "min": 55.29,
      "max": 74.48,
      "night": 56.93,
      "eve": 68.86,
      "morn": 55.29
    },
    "pressure": 987.88,
    "humidity": 72,
    "weather": [
      {
        "id": 800,
        "main": "Clear",
        "description": "clear sky",
        "icon": "01d"
      }
    ],
    "speed": 5.61,
    "deg": 333,
    "clouds": 0
```

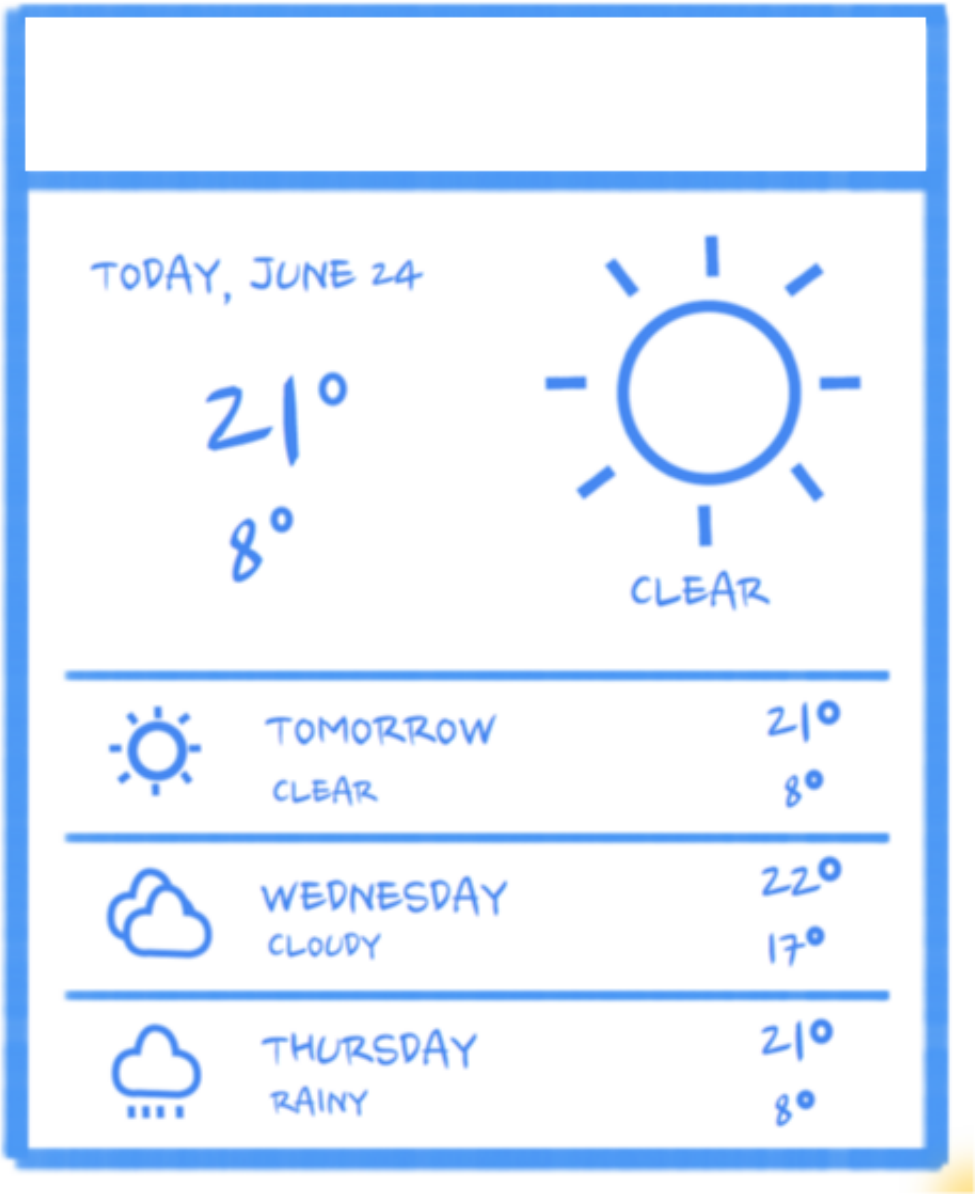
```
    },
    {
      "dt": 1461517200,
      "temp": {
        "day": 75.72,
        "min": 54.55,
        "max": 77.67,
        "night": 58.35,
        "eve": 72.32,
        "morn": 54.55
      },
      "pressure": 990.33,
      "humidity": 56,
      "weather": [
        {
          "id": 800,
          "main": "Clear",
          "description": "clear sky",
          "icon": "01d"
        }
      ],
      "speed": 3.71,
      "deg": 143,
      "clouds": 0
    },
    {
      "dt": 1461603600,
      "temp": {
        "day": 78.4,
        "min": 65.95,
        "max": 78.4,
        "night": 67.08,
        "eve": 76.19,
        "morn": 65.95
      },
      "pressure": 989.21,
      "humidity": 0,
      "weather": [
        {
          "id": 500,
          "main": "Rain",
          "description": "light rain",
```

```
          "icon": "10d"
        }
      ],
      "speed": 8.5,
      "deg": 194,
      "clouds": 37,
      "rain": 1.3
    },
    {
      "dt": 1461690000,
      "temp": {
        "day": 82.9,
        "min": 65.89,
        "max": 82.9,
        "night": 71.83,
        "eve": 82.81,
        "morn": 65.89
      },
      "pressure": 986.97,
      "humidity": 0,
      "weather": [
        {
          "id": 800,
          "main": "Clear",
          "description": "clear sky",
          "icon": "01d"
        }
      ],
      "speed": 7.45,
      "deg": 240,
      "clouds": 10
    }
  ]
}
```


NOTE: A definition of each field can be found under the ***Weather parameters in API respond*** section here: http://openweathermap.org/forecast5

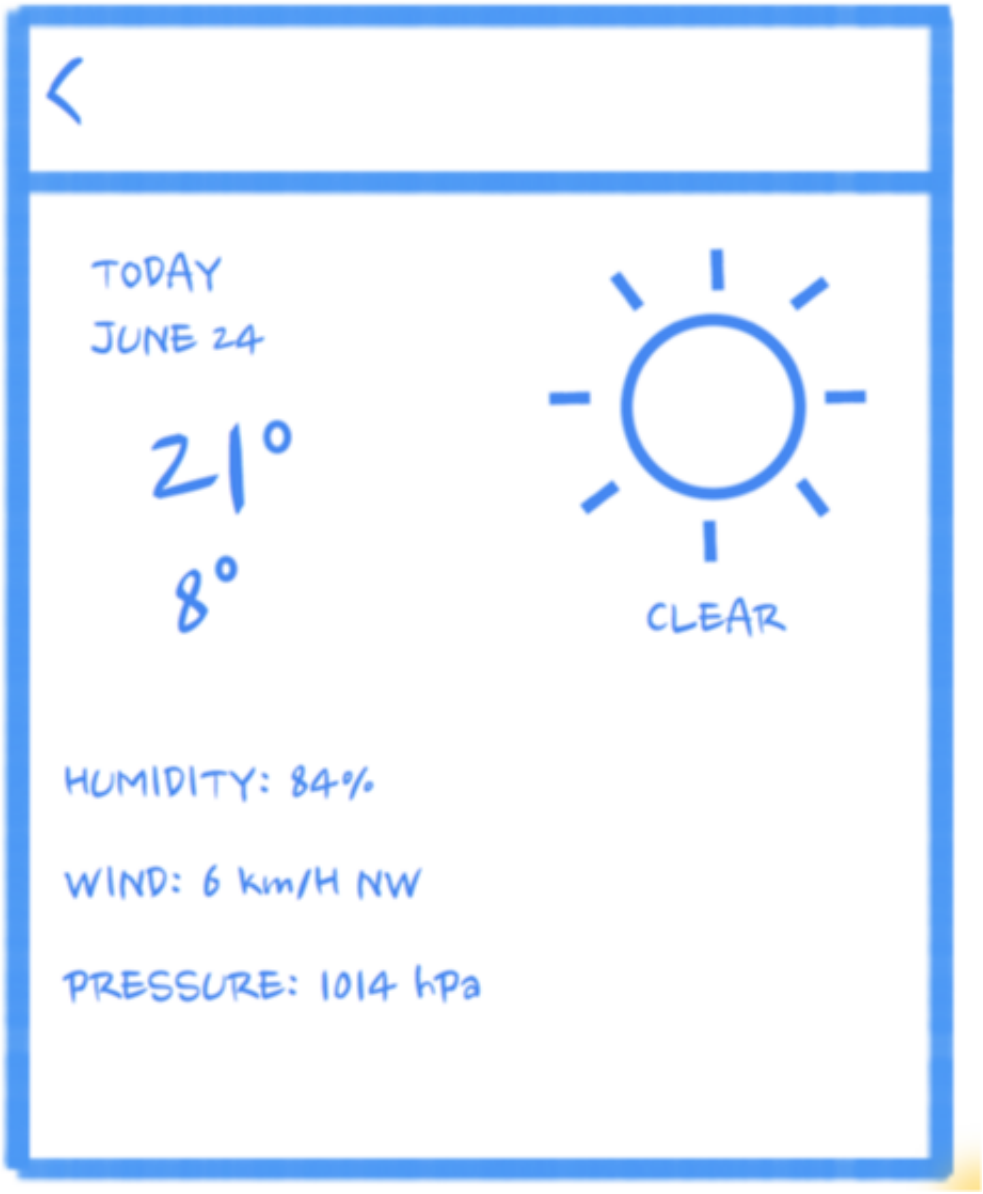
Wireframes

Main View



- The *Today, June 24* portion uses the data retrieved when getting the Current Weather from the data service
- *Tomorrow, Wednesday, and Thursday* portions uses the data retrieved when getting the 5-day forecast

Detail View



- The Detail View can be accessed when the user touches any item - whether it be the current day's weather or a day in the forecast.

Mockups

Main View

Today, June 19

20°

13°

Clear

Tomorrow	Clouds	18° / 12°
Saturday	Rain	14° / 13°
Sunday	Clear	14° / 14°
Monday	Rain	14° / 14°
Tuesday	Clear	14° / 14°

Main View shows Current Weather and multi-day forecast in list below.

Detail View

Tomorrow

June 19

18°

12°

Clear

Humidity: 81 %

Pressure: 1017 hPa

Wind: 2 km/h NW

Detail View is the detail view for either the current or forecasted weather when selected by the user

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Images

You will see in the responses from the data service, a field named, ***icon***. For example, the icon for the Current Weather call above is ***01n***.

Use this chart to map the field value to the image file name

01d	01n	art_clear.png	ic_clear.png		
02d	02n	art_light_clouds.png	ic_light_clouds.png		
03d	03n	art_clouds.png	ic_cloudy.png		
04d	04n	art_clouds.png	ic_cloudy.png		
09d	09n	art_light_rain.png	ic_light_rain.png		
10d	10n	art_rain.png	ic_rain.png		
11d	11n	art_storm.png	ic_storm.png		
13d	13n	art_snow.png	ic_snow.png		
50d	50n	art_fog.png	ic_fog.png		

NOTE: The images in the grid can be found in the zip file that accompanies this PDF.